CSE/ISyE 6740
Computational Data Analysis

Diffusion Models

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Kai Wang, Assistant Professor in Computational Science and Engineering
kwang692@gatech.edu

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Outline

- **Advanced Topics**
  - Diffusion Models
    - Diffusion process
    - Denoising process
    - DDPM training loss function
    - Diffusion model applications

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From Autoencoder to Diffusion Models

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Autoencoders v.s. Variational Autoencoders

- **AE loss**:
  - Encoder $z = q(\cdot\,|x, \phi)$
  - Decoder $\mathrm{x}' = p(\cdot\,|z, \theta)$

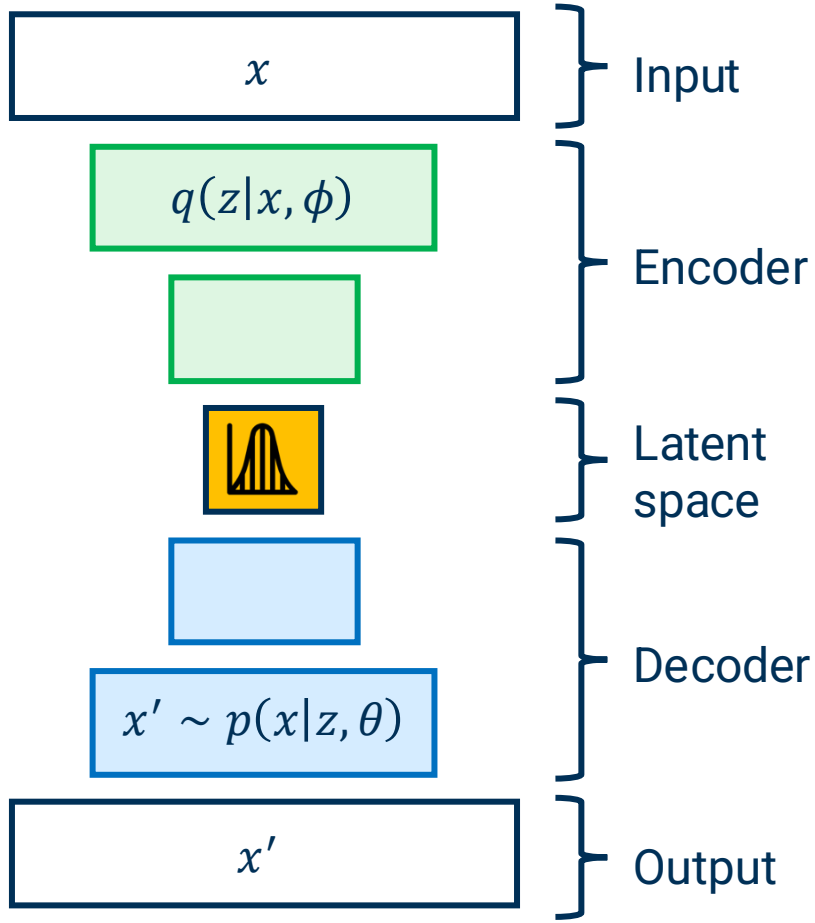
$$\min_{\phi,\theta} \quad \|x - x'\|^2 \quad \text{(recon. loss)}$$

$$\text{where } z = q(\cdot\,|x, \phi),\ x' = p(\cdot\,|z, \theta)$$

- **VAE loss**:
  - Encoder $z \sim q(\cdot\,|x, \phi) = \mathcal{N}\left(\mu_\phi(x_i), \sigma^2_\phi(x_i)\right)$
  - Decoder $\mathrm{x}' \sim p(\cdot\,|z, \theta) = \mathcal{N}(G_\theta(z), \eta I)$

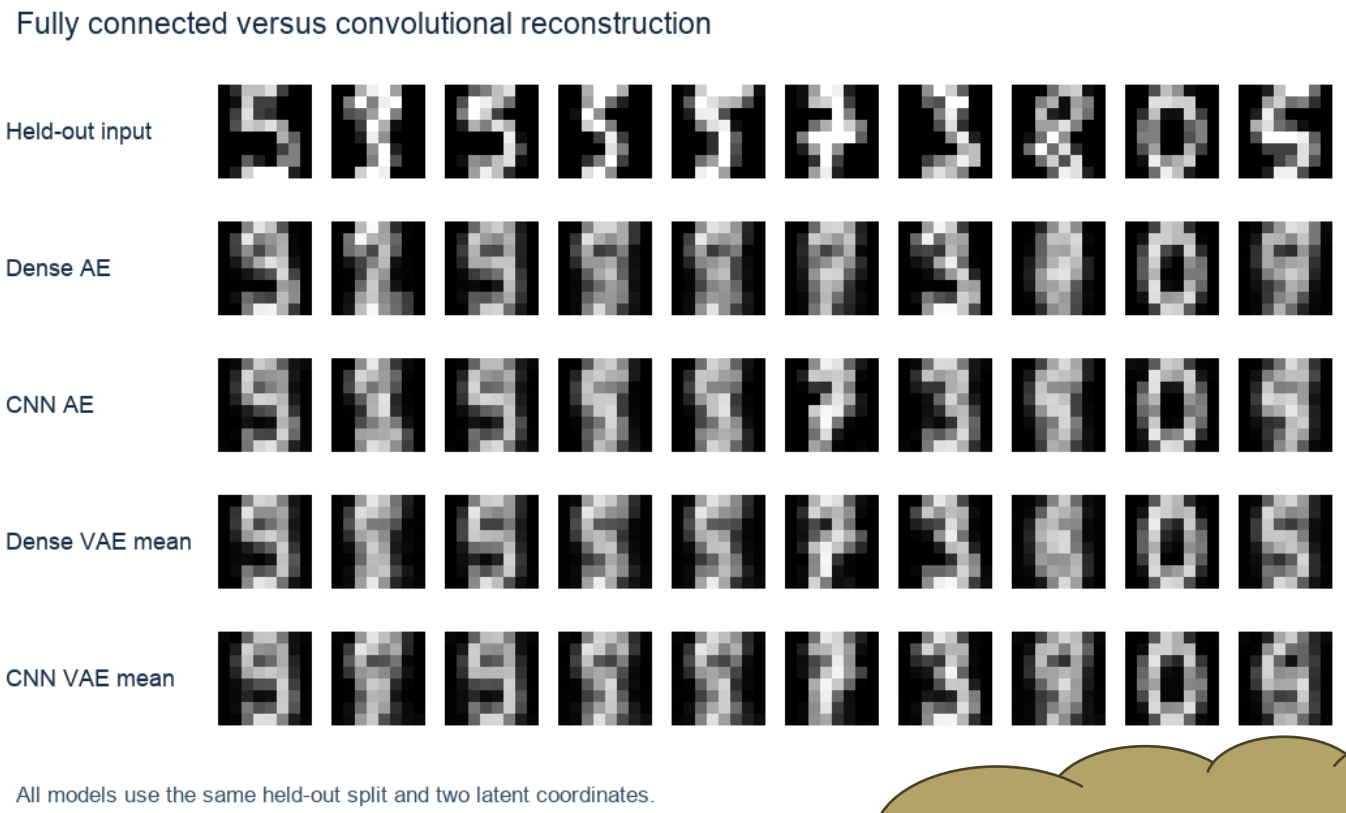
$$\min_{\phi,\theta} \quad \|x - x'\|^2 + D_{KL}\left(q||\mathcal{N}(0,1)\right) \text{ (recon. + KL)}$$

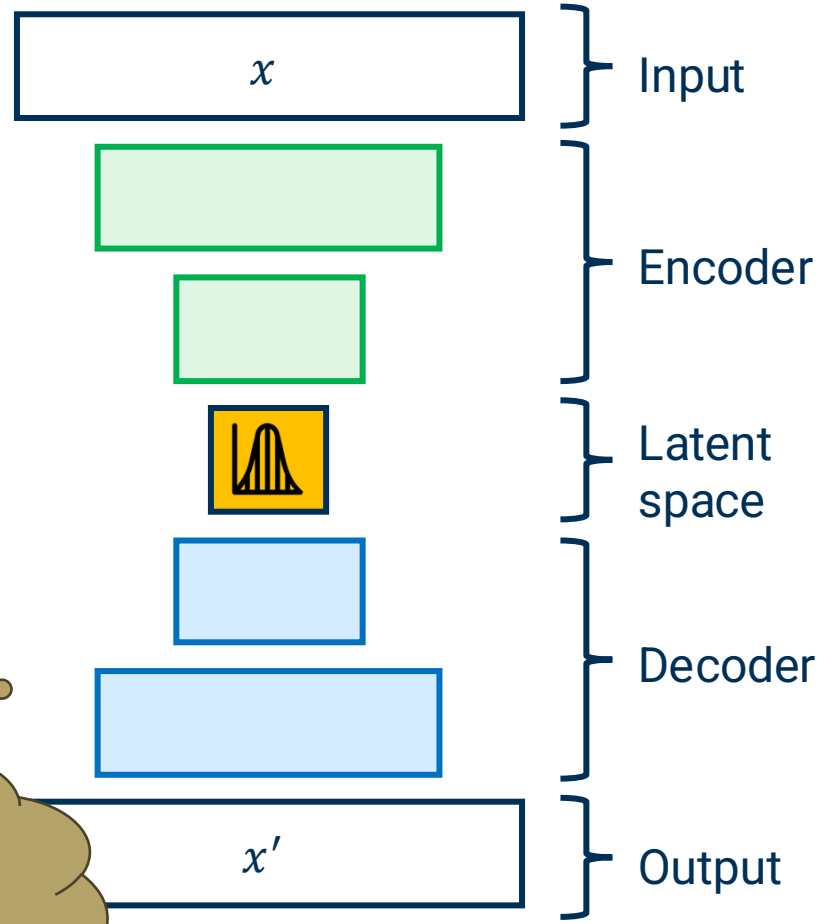
$$\text{where } z \sim q(\cdot\,|x, \phi),\ x' \sim p(\cdot\,|z, \theta)$$



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Autoencoders v.s. Variational Autoencoders

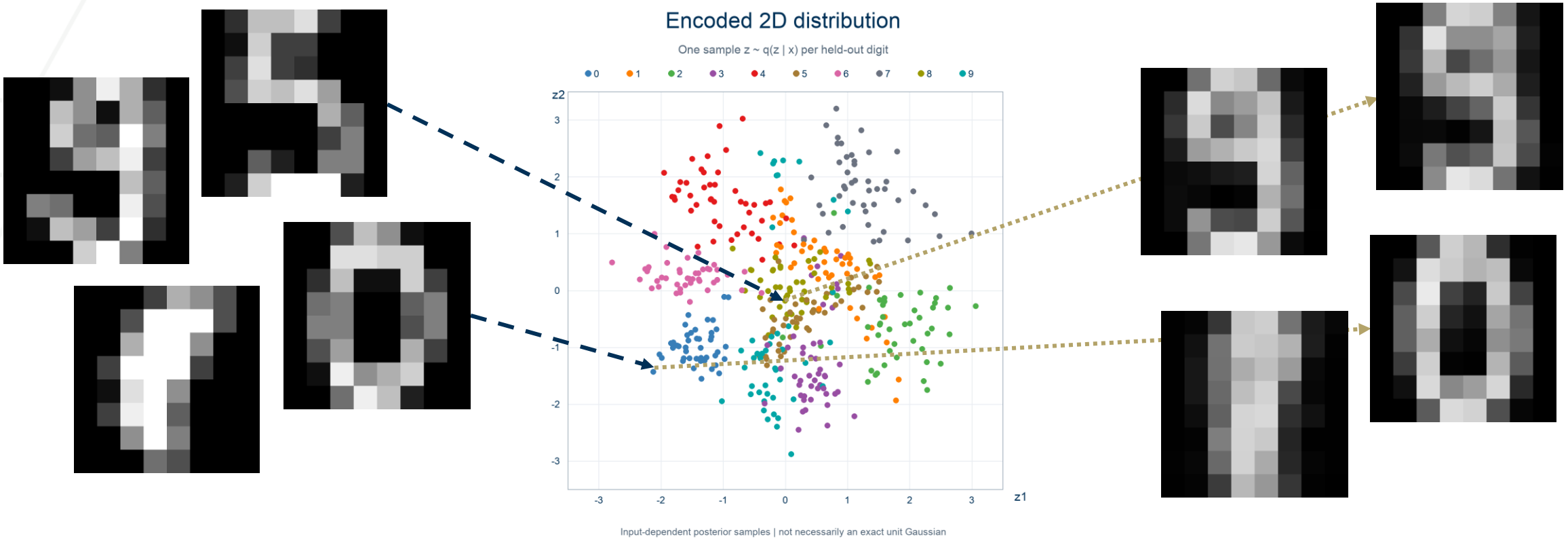




What are the potential issues of VAEs?

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Take a Closer Look at VAEs



Input images (8x8)

Latent variable 2D

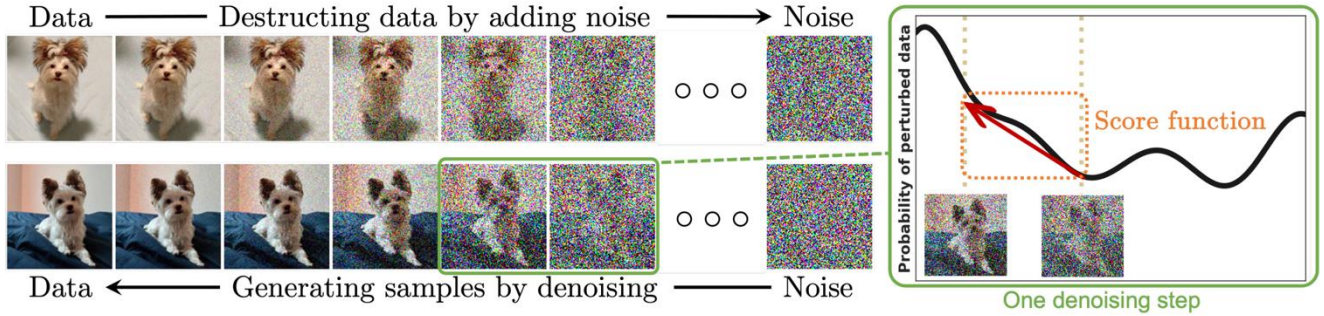
Output images (8x8)

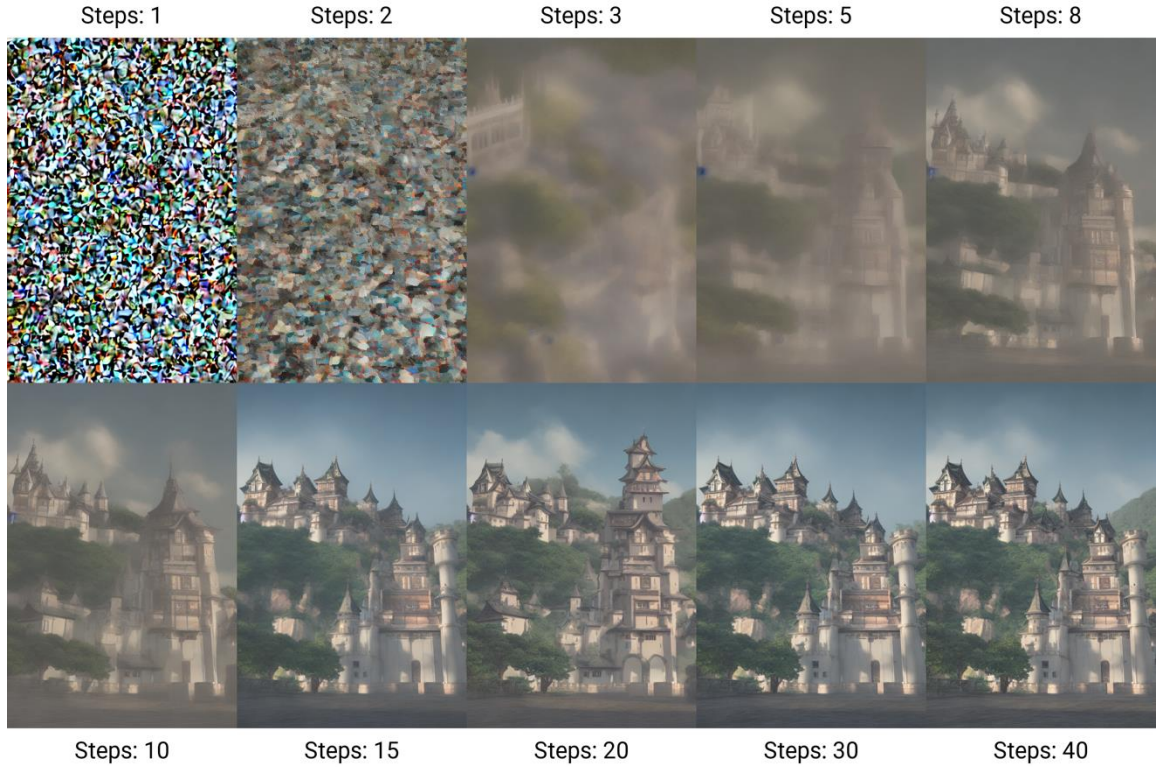
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Diffusion Models

- Diffusion model was proposed in 2020 by [Ho et al. 2020](Ho et al. 2020).
  - **Diffusion process**: gradually adding noise to an image until it becomes completely noise.
  - **Denoising process**: to generate an image, start from a random noise and gradually denoise it until it becomes clear.

- An extension of VAEs



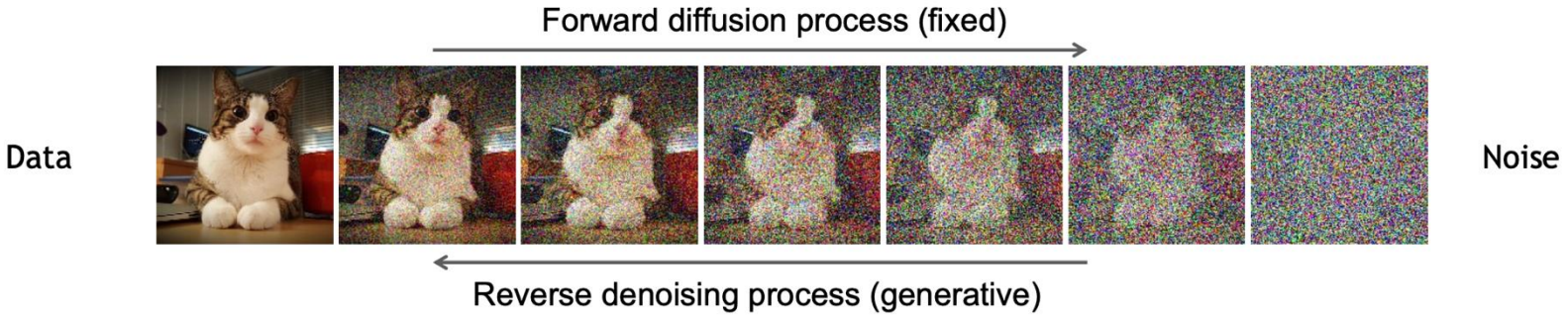


Diffusion Process

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Diffusion Process

- Can we gradually turn an image to a noise, and gradually turn a noise back to the image?



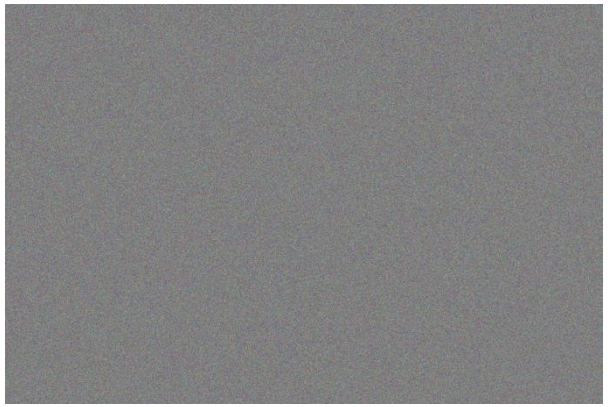
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Diffusion Process

- Add noise to an image



$\times\sqrt{1-\beta_t}$



$\times\sqrt{\beta_t}$

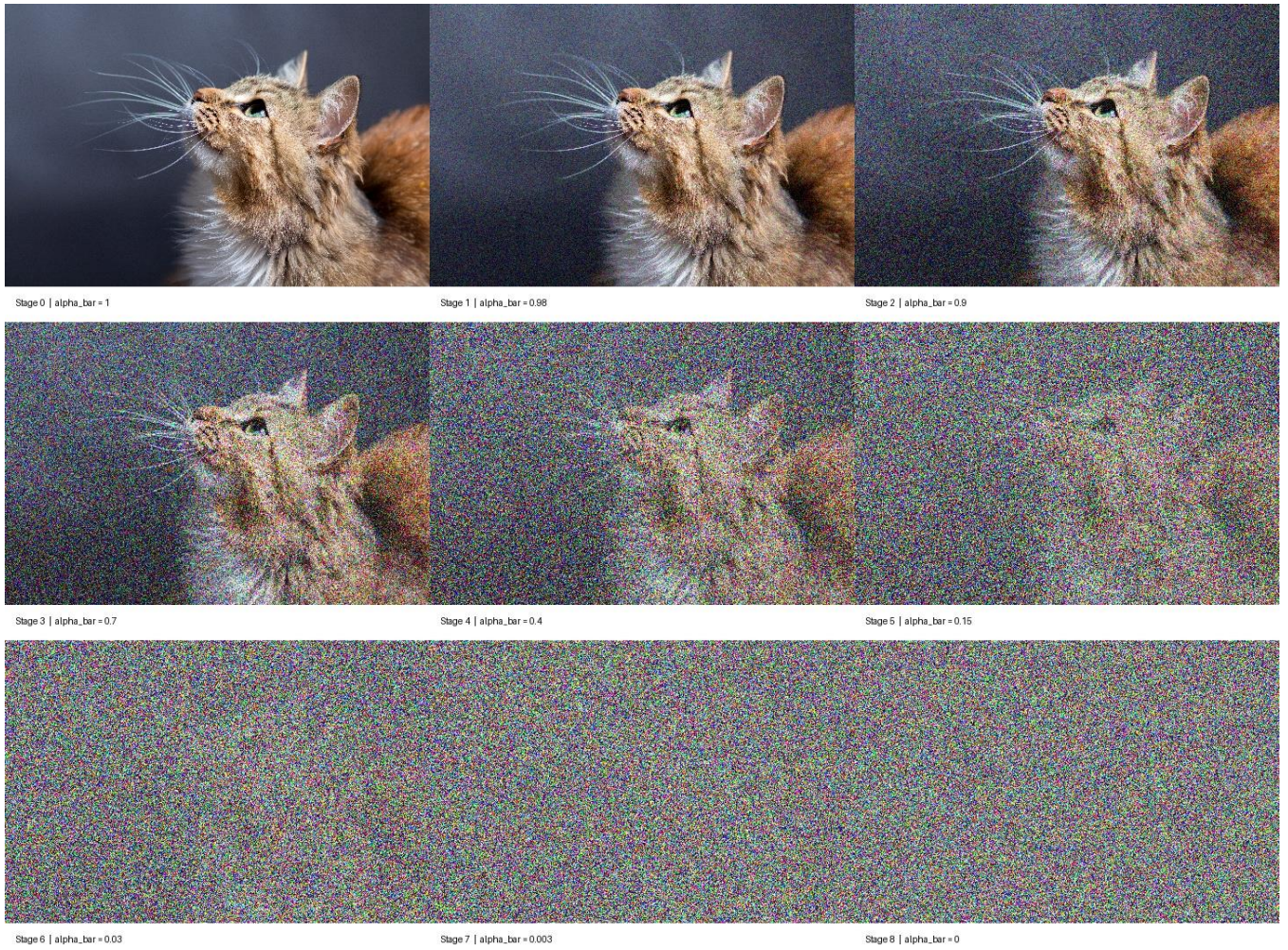


x_{t-1} $\mathcal{N}(\mathbf{0},\mathbf{1}) \in \mathbb{R}^{d_x}$ x_t

Question: why square root?

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Diffusion Process



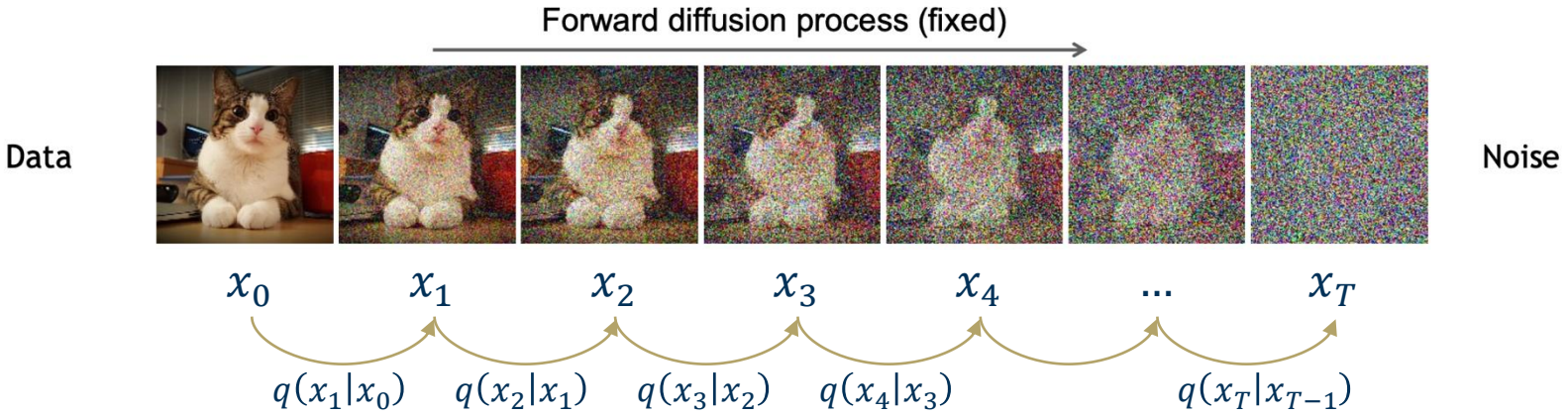
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Diffusion Process

- The formal definition of the forward process in T steps:

$$q(x_t|x_{t-1}) = \mathcal{N}\left(x_t; \sqrt{1-\beta_t}x_{t-1}, \beta_t I\right), \qquad q(x_{1:T} \mid x_0) = \prod_{t=1}^{T} q(x_t|x_{t-1})$$

- Parameters β_t are the noise schedule, which gradually add noise to the input such that $q(x_T \mid x_0) \approx \mathcal{N}(x_T; 0, I)$



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Diffusion Process

- Signal from image at time t

$$x_0 \times \sqrt{1-\beta_1} \times \sqrt{1-\beta_2} \times \cdots \times \sqrt{1-\beta_t} = \left(\prod_{s=1}^{t} \sqrt{1-\beta_s} \right) x_0 = \sqrt{\bar{\alpha}_t}\, x_0$$

- Cumulative noise at time t

$$\sqrt{1-\bar{\alpha}_t}\,\epsilon \qquad where\ \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{1})$$

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon$$

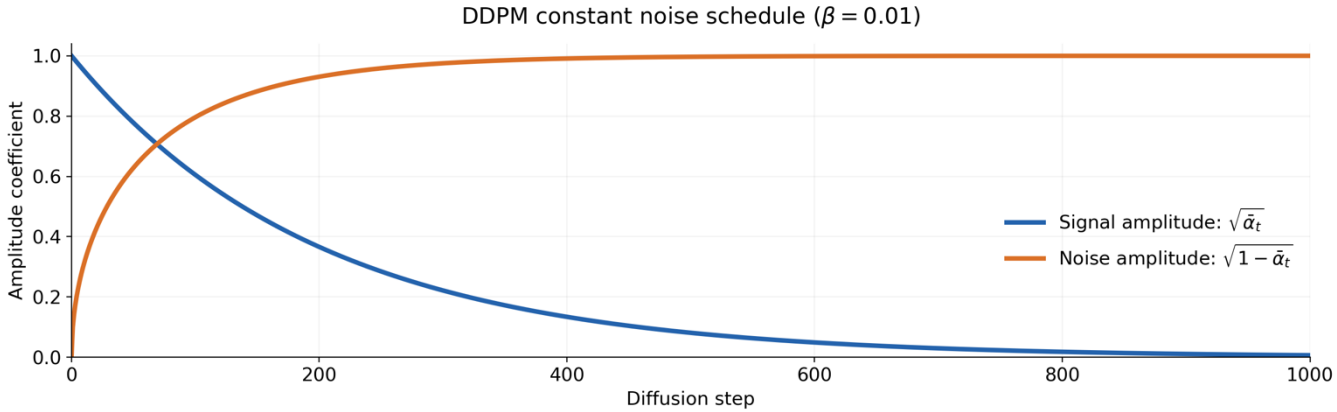
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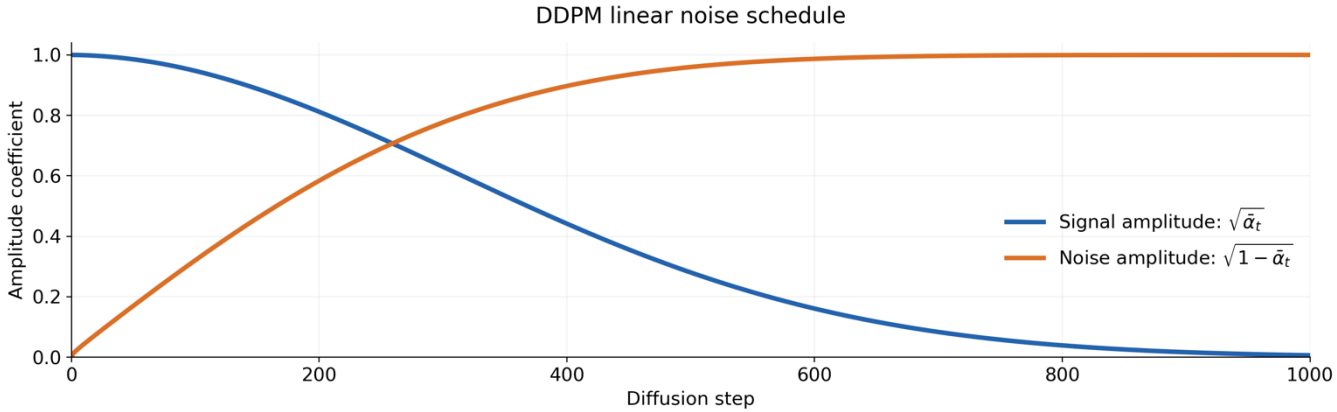
Noise Scheduler

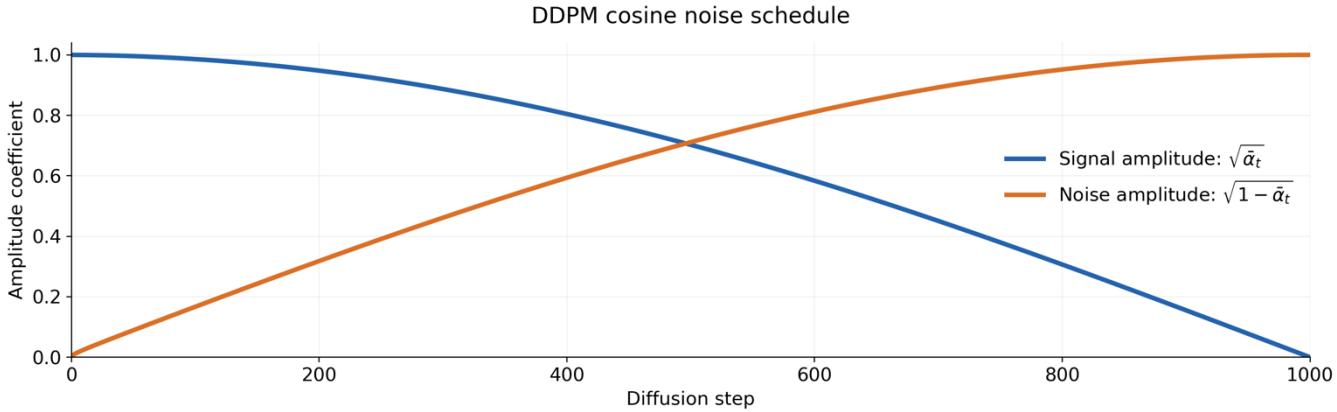
- How to choose the noise size β_t at step t?
  - **Constant noise scheduler:** $\beta_t = 0.01$
  - **Linear noise scheduler**: $\beta_1 = 10^{-4}$ linearly increases to $\beta^T = 0.02$
  - **Cosine noise scheduler**:

$$\beta_t = clip\left(1 - \frac{\bar{\alpha}_t}{\bar{\alpha}_{t-1}}, 0.999\right), \bar{\alpha}_t = \frac{f(t)}{f(0)}$$

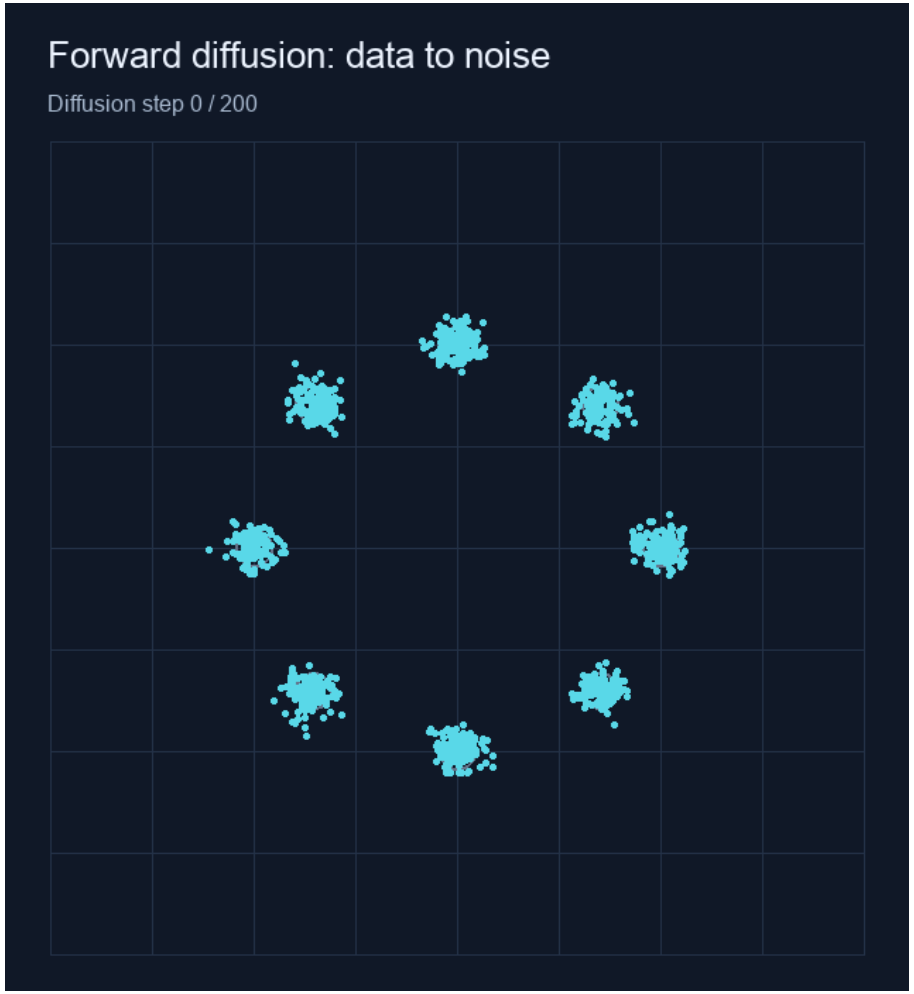
$$where\ f(t) = \cos\left(\frac{\frac{t}{T} + s}{1 + s} \cdot \frac{\pi}{2}\right)^2$$

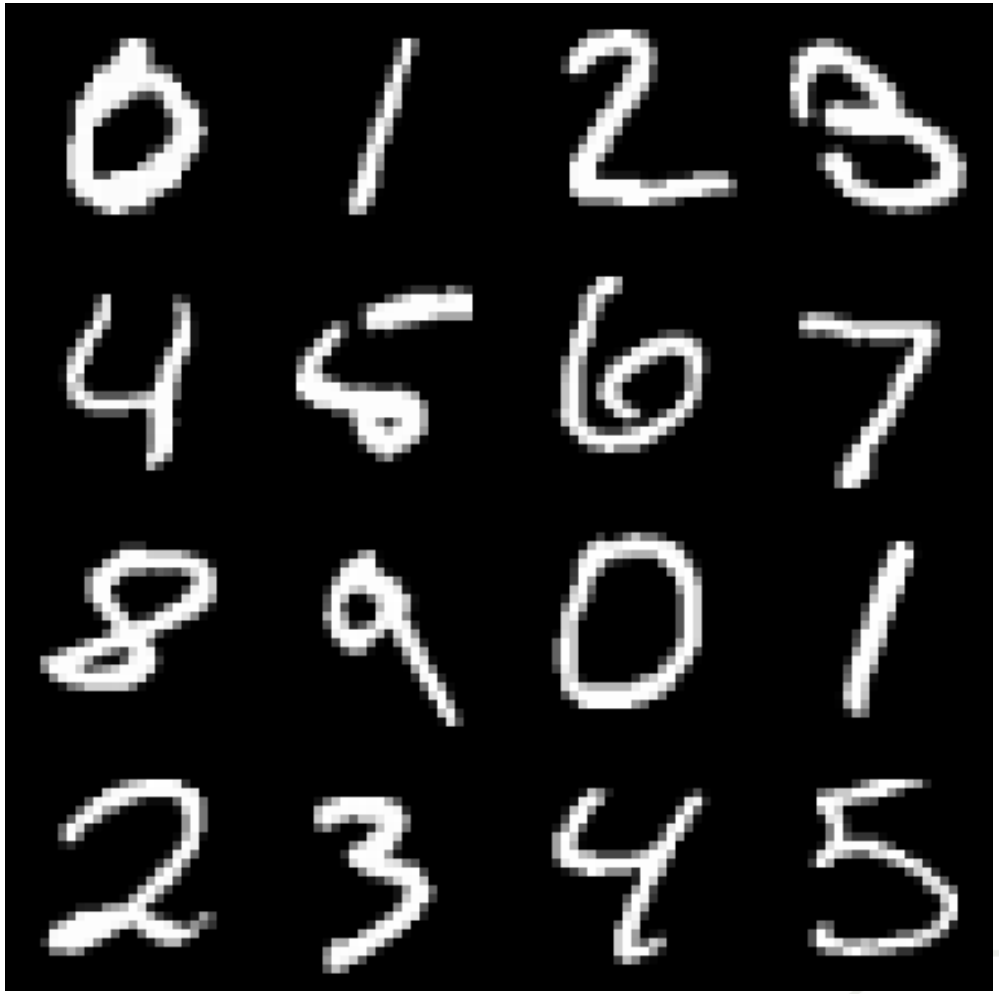






Diffusion Process





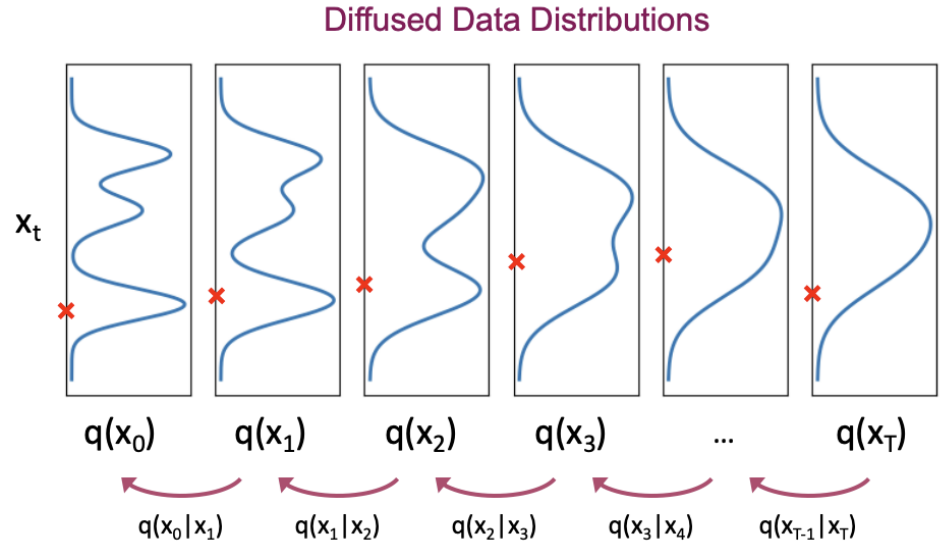
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Denoising Process

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Denoising Process

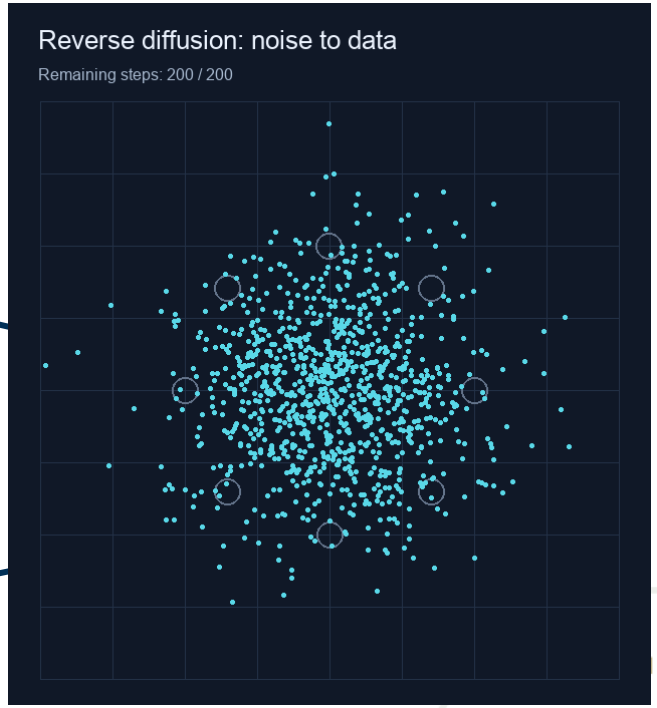
- Recall that the diffusion parameters are designed such that $q(x_T \mid x_0) \approx \mathcal{N}(x_T; 0, I)$



In the denoising process:

- Sample $x_T \sim \mathcal{N}(x_T; 0, I)$
- Iteratively sample $x_{t-1} \sim q(x_{t-1}|x_t)$

We only know how to mess up in $q(x_t|x_{t-1})$.
It is hard to tidy up $q(x_{t-1}|x_t)$…

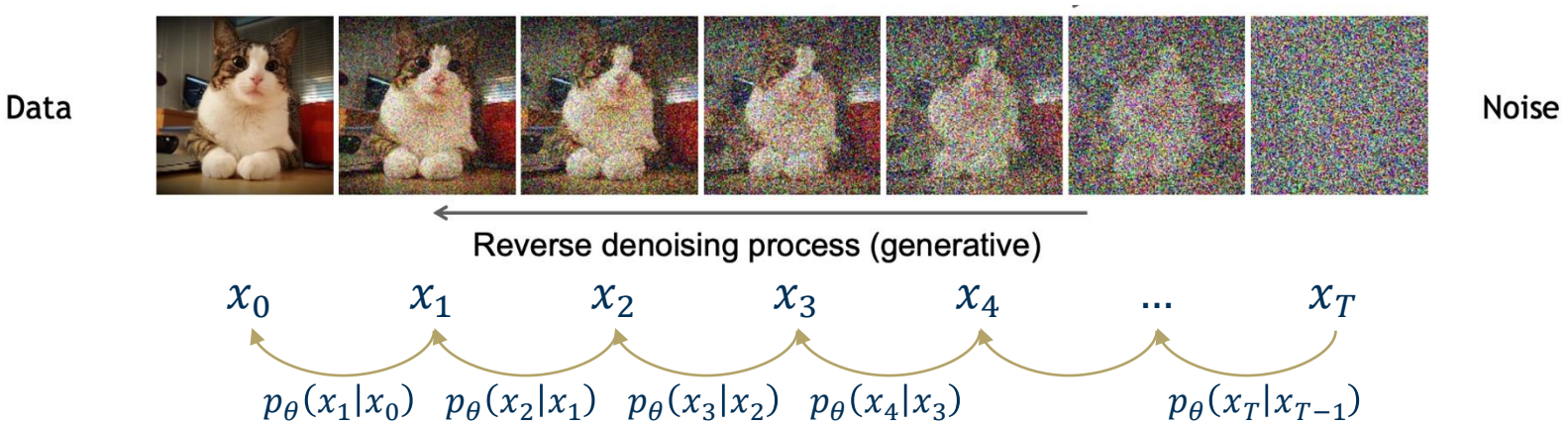


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Denoising Process

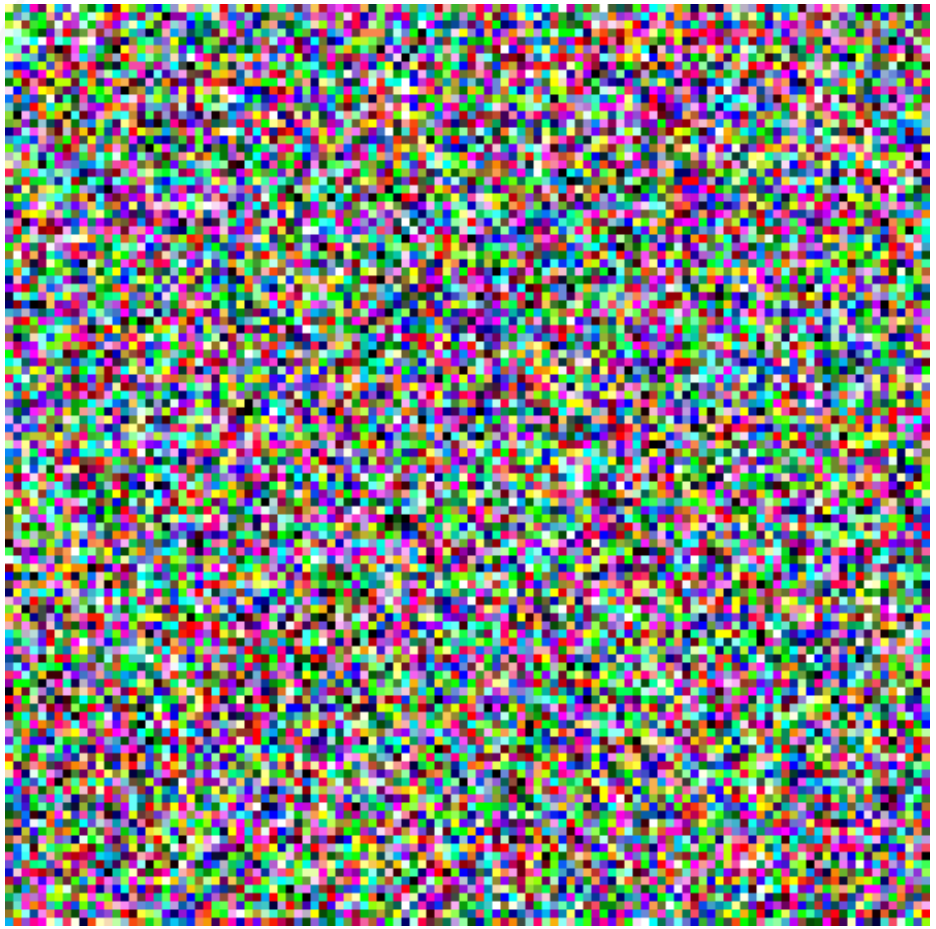
- **Solution**: similar to VAE, just use a Gaussian to approximate $q(x_{t-1}|x_t)$

$$q(x_{t-1}|x_t) \approx p_\theta(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t), \sigma_t^2 I)$$

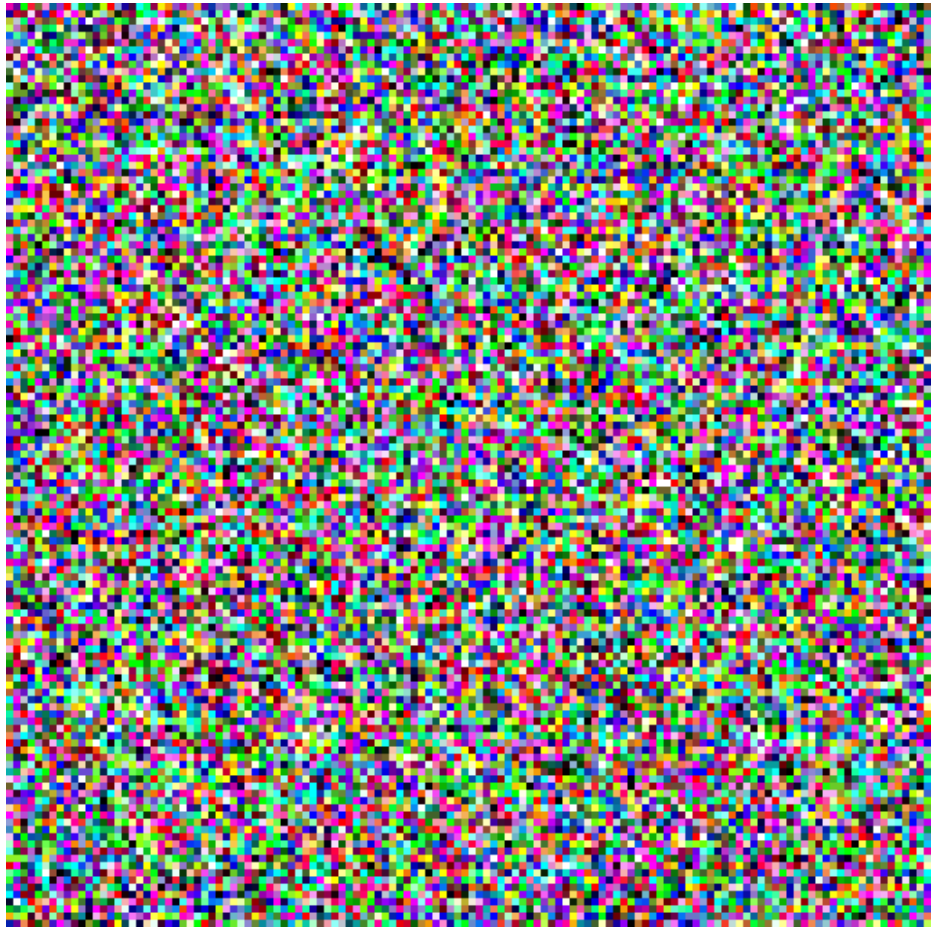


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Denoising Process



MNIST



ImageNet

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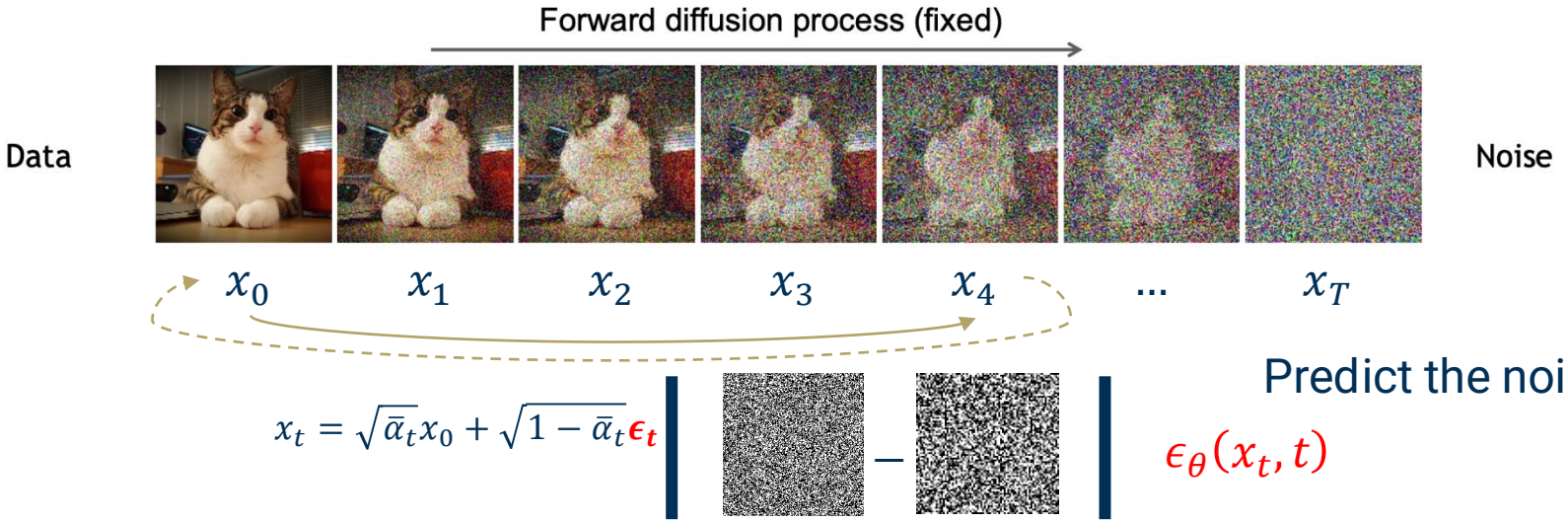
DDPM Loss

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Noise Model

- How to train the model?
  - Train a **noise model $\epsilon_\theta(x_t, t)$** to predict how much noise to deduct from the corrupted image/sample x_t.

$$L_{DDPM} = \sum_t \mathbb{E}_{t\sim[1,T],\, x_0\sim D_{train},\, \epsilon_t\sim\mathcal{N}(0,1)} \left[\|\epsilon_t - \epsilon_\theta(x_t, t)\|^2\right]$$



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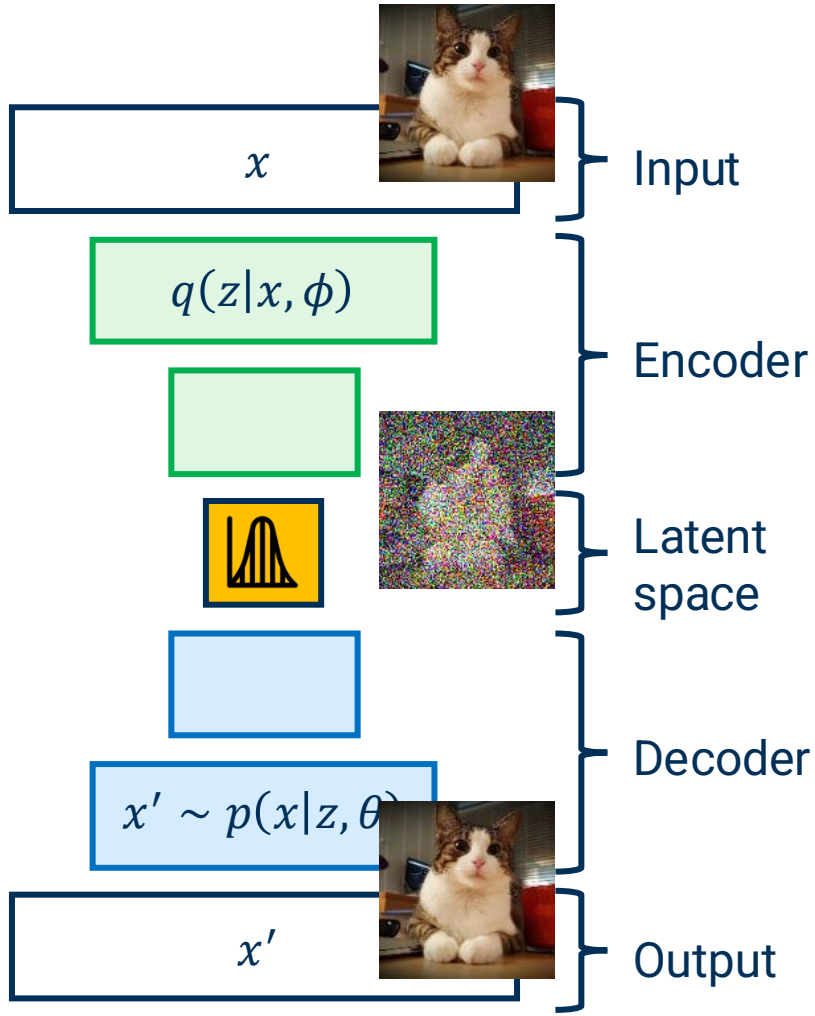
Noise Model

- The DDPM loss is similar to the **reconstruction loss** $\|x - x'\|^2$ in VAEs.
  - Input $x = x_0$
  - Latent/corrupted variable $z = x_t$
  - Decode back to get the output x'

- Encoder is the diffusion process at time t
  - $x_t = \sqrt{\bar{\alpha}_t}x + \sqrt{1 - \bar{\alpha}_t}\epsilon$

- Decoder is parameterized as
  - $x' = \frac{x_t - \sqrt{1-\bar{\alpha}_t}\,\epsilon_\theta(x_t, t)}{\sqrt{\bar{\alpha}_t}}$.

$$\|x - x'\|^2 = \frac{1 - \bar{\alpha}_t}{\bar{\alpha}_t}\|\epsilon - \epsilon_\theta(x_t, t)\|^2$$

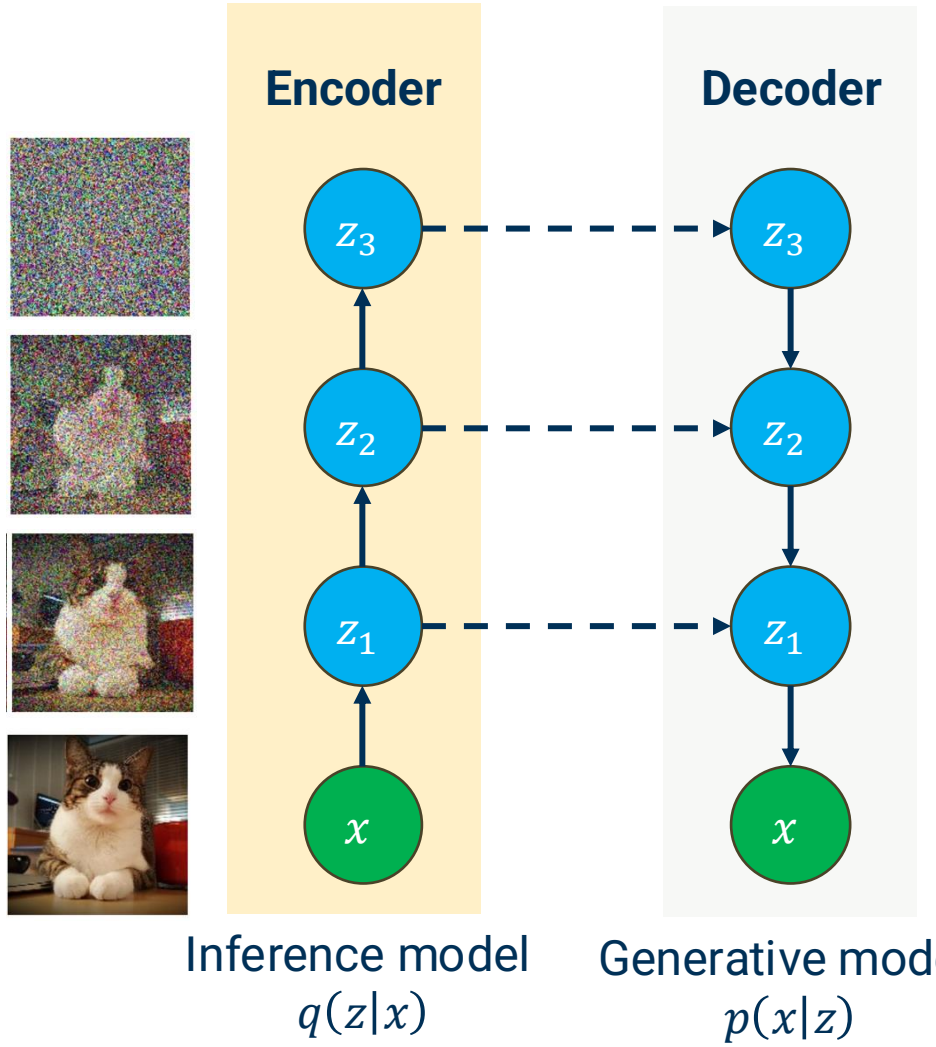


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Connection to VAE

- Diffusion models can be considered as a special form of **hierarchical VAEs**

- In diffusion models:
  - The encoder is fixed
  - The latent variables have the same dimension as the data
  - The loss is decomposed to each time step
  - The model is trained with a weighted ELBO



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DDPM Algorithm

Denoising Diffusion Probabilistic Models

The loss function can be written as:

$$L_{DDPM} = \sum_t \mathbb{E}_{t\sim[1,T],\, x_0\sim D_{train},\, \epsilon_t\sim\mathcal{N}(0,1)} \left[\|\epsilon_t - \epsilon_\theta(x_t, t)\|^2\right]$$

Algorithm 1 Training

1: **repeat**
2: $\mathbf{x}_0 \sim q(\mathbf{x}_0)$
3: $t \sim \text{Uniform}(\{1, \ldots, T\})$
4: $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
5: Take gradient descent step on
$$\nabla_\theta \left\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, t)\right\|^2$$
6: **until** converged

Algorithm 2 Sampling

1: $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
2: **for** $t = T, \ldots, 1$ **do**
3: $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ if $t > 1$, else $\mathbf{z} = \mathbf{0}$
4: $\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\right) + \sigma_t \mathbf{z}$
5: **end for**
6: **return** $\mathbf{x}_0$

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Implementation Consideration

- Use a neural network to represent $\epsilon_\theta(x_t, t)$
  - Noisy input x_t
  - Time representation of t

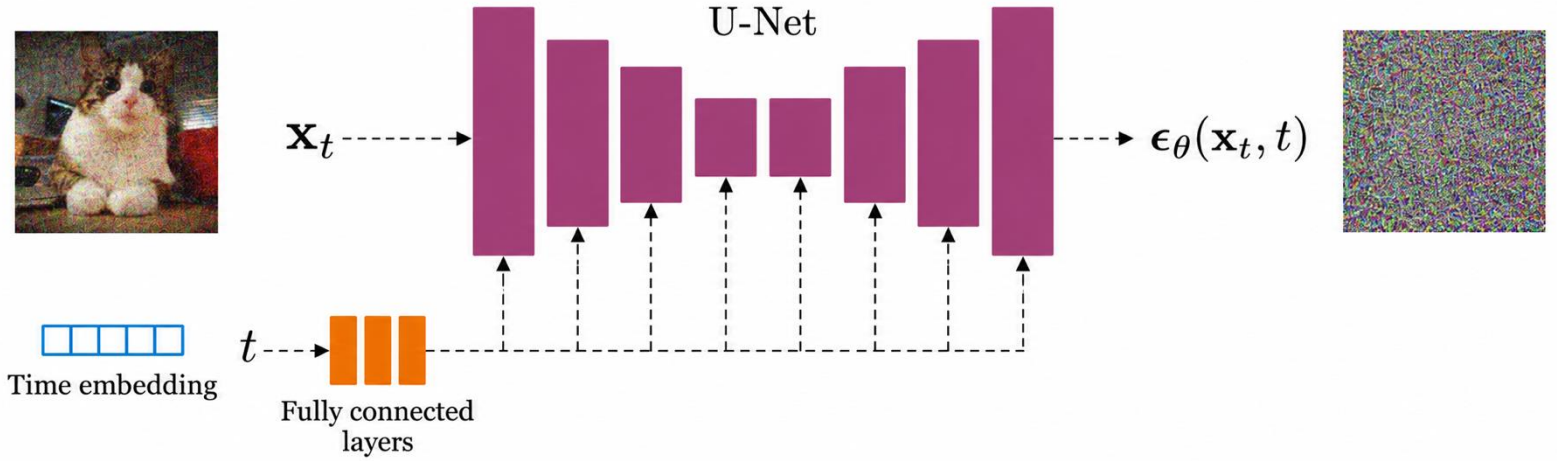
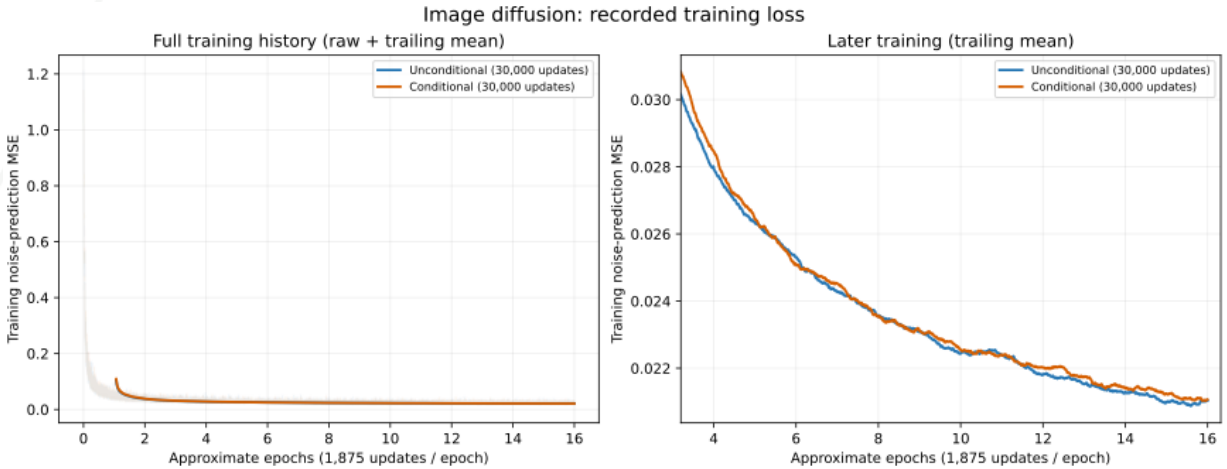
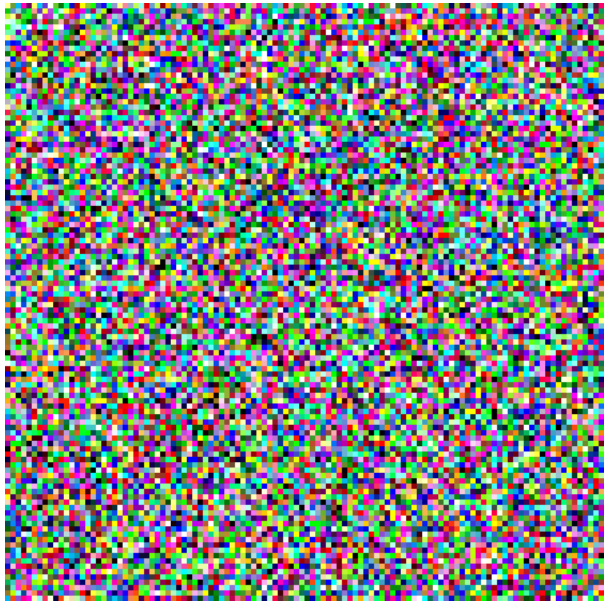


Image credit https://cvpr2022-tutorial-diffusion-models.github.io/

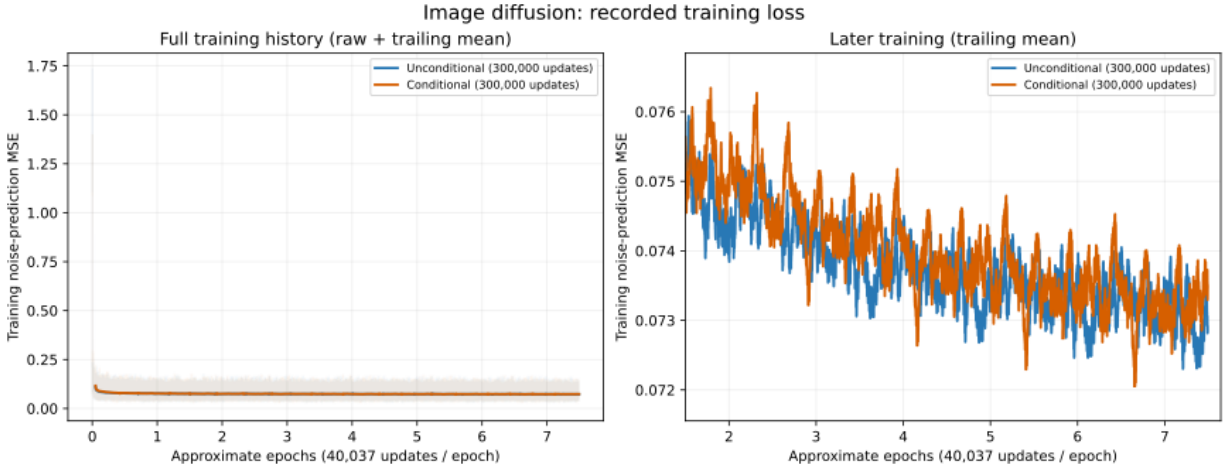
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Training Demos





MNIST (60000 samples, 28x28, 10 classes, 55MB)





ImageNet (1.28M samples, varying size -> 32x32x3, 1000 classes, 138GB)

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DDPM Loss Derivation (Homework 3)

- Similarly, we can compute the log-likelihood of receiving the same input x as output

$$\mathbb{E}_{x_0 \sim D_{train}}[-\log p(x_0)] \leq \mathbb{E}_{x_0 \sim D_{train}} \mathbb{E}_{q(x_{1:T}|x_0)} \left[-\log \frac{p_\theta(x_{0:T})}{q(x_{1:T}|x_0)} \right] := L$$

Jensen inequality

$$L = \mathbb{E}_{x_0 \sim q} \left[\underbrace{D_{KL}(q(x_T|x_0)||p(x_T))}_{L_T} + \sum_{t=2}^{T} \underbrace{D_{KL}(q(x_{t-1}|x_t, x_0)||p_\theta(x_{t-1}|x_t))}_{L_{t-1}} \underbrace{- \log p_\theta(x_0|x_1)}_{L_0} \right]$$

L_T: $\approx$ constant

L_{t-1}

L_0: Ignored

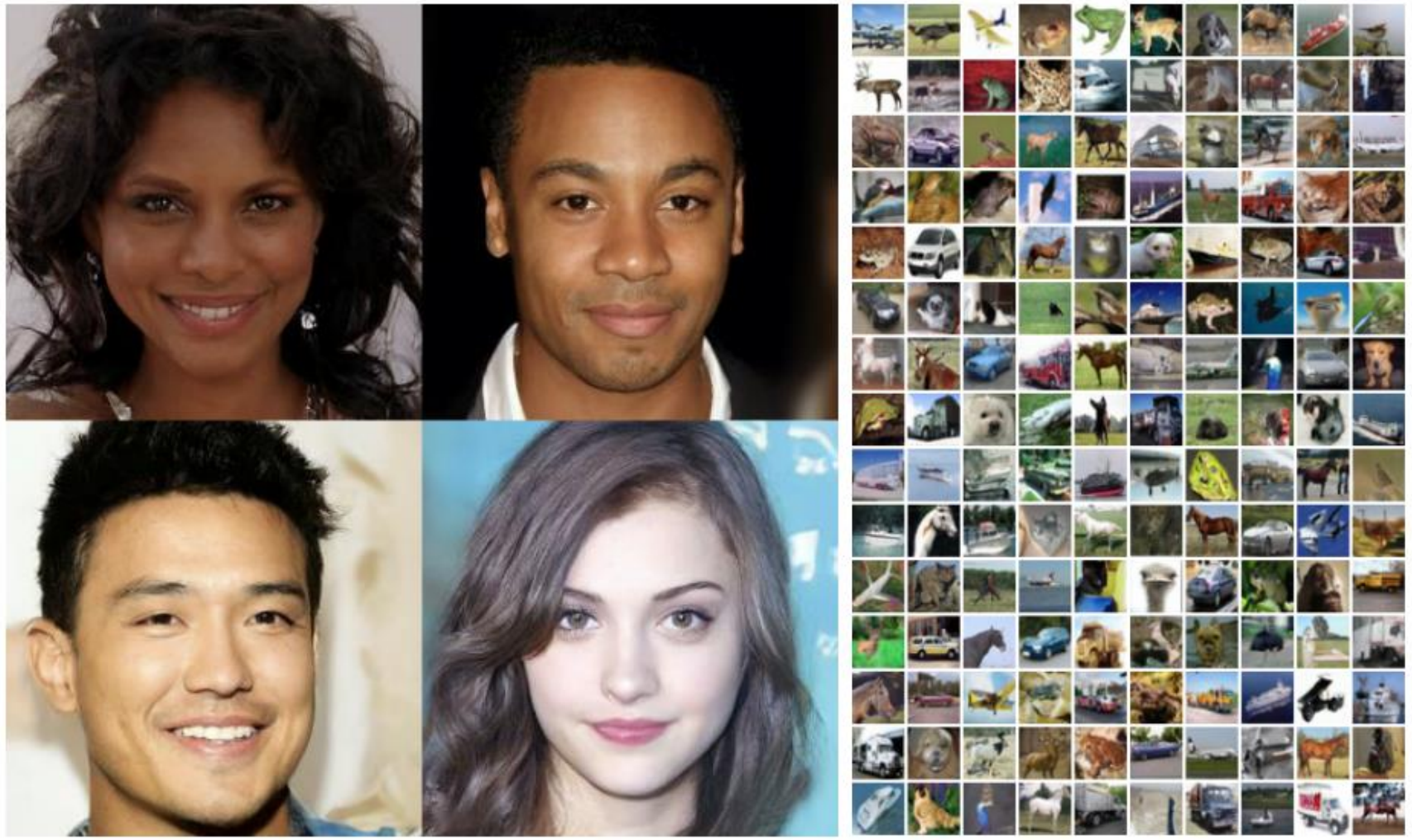
Please read https://lilianweng.github.io/posts/2021-07-11-diffusion-models/ to understand how the DDPM derivation works and finish your homework 3.

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Diffusion Model Application

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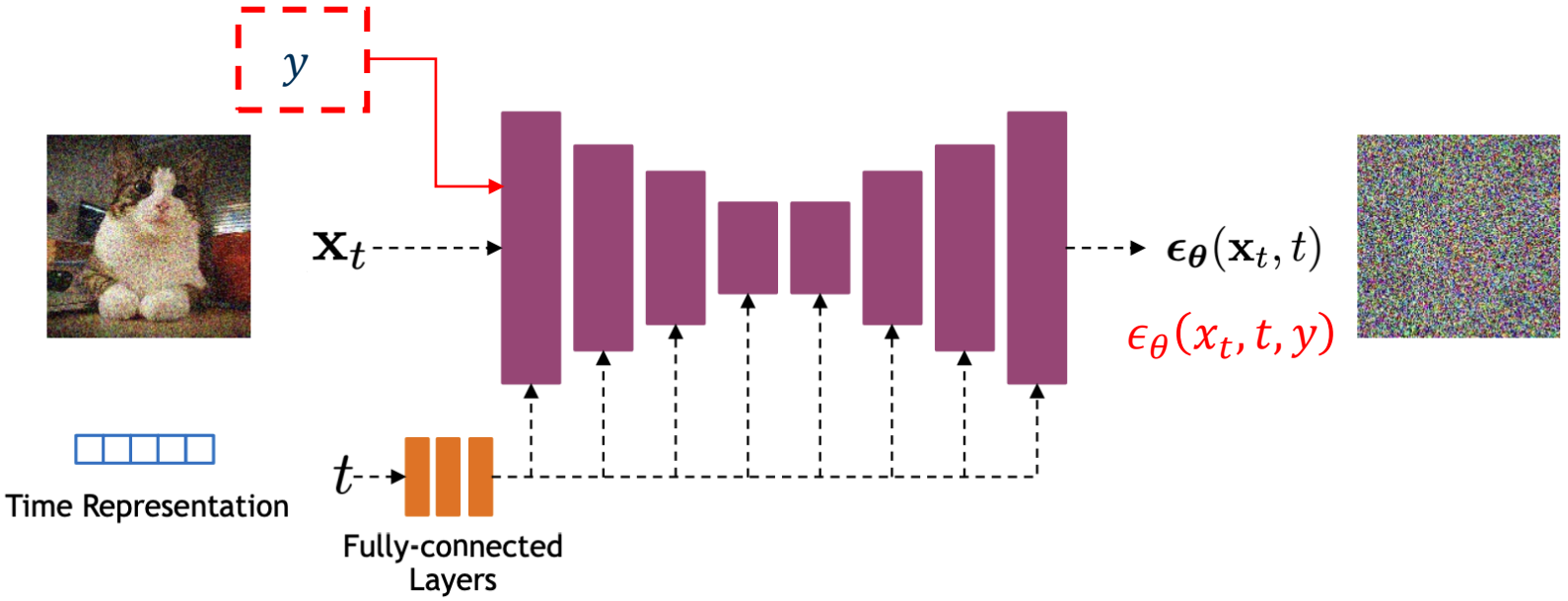
Diffusion Model Applications



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Conditional Diffusion Model

- Can we also condition on an additional class/context information y?
- Use a neural network to represent $\epsilon_\theta(x_t, t)$
  - Noisy input x_t
  - Time representation of t
  - Context y

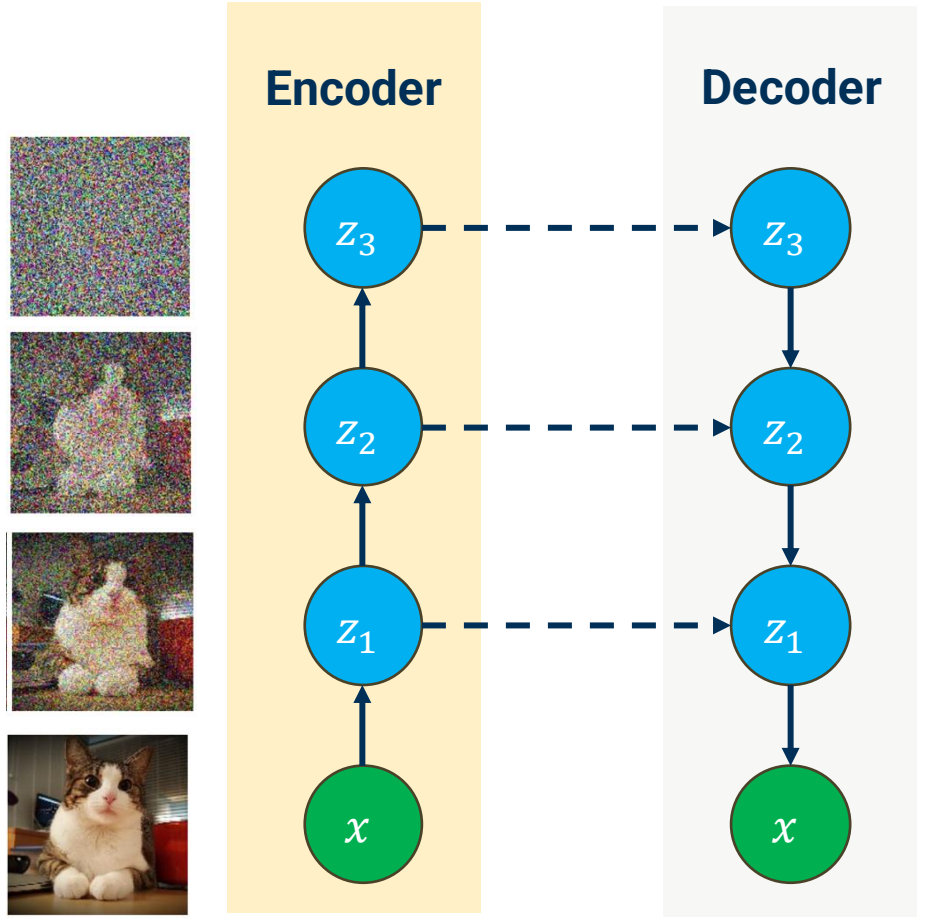


Slides based on https://cvpr2022-tutorial-diffusion-models.github.io/

Pros and Cons of VAE and Diffusion Models?

- **VAE**
  - 👑 Stronger latent space representation
  - 👑 Can be used to obtain embedding
  - 👑 Less compute needed
  - More unstable
  - Lower resolution

- **Diffusion models**
  - Simpler latent space representation
  - No encoder learned
  - More compute needed
  - 👑 More stable
  - 👑 Higher resolution



Inference model
$q(z|x)$

Generative model
$p(x|z)$

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Applications of VAE and Diffusion Models?

- We see so many fancy applications of VAE and diffusion models. But can we use them to do something else?
- Technically, these models are trying to do **density estimation**
  - Try to estimate $p(x)$ and $p(x \mid y)$ from a dataset D
- We have parametric models (Gaussian, GMM etc), but VAE and diffusion models offer a **deep learning option** for us.

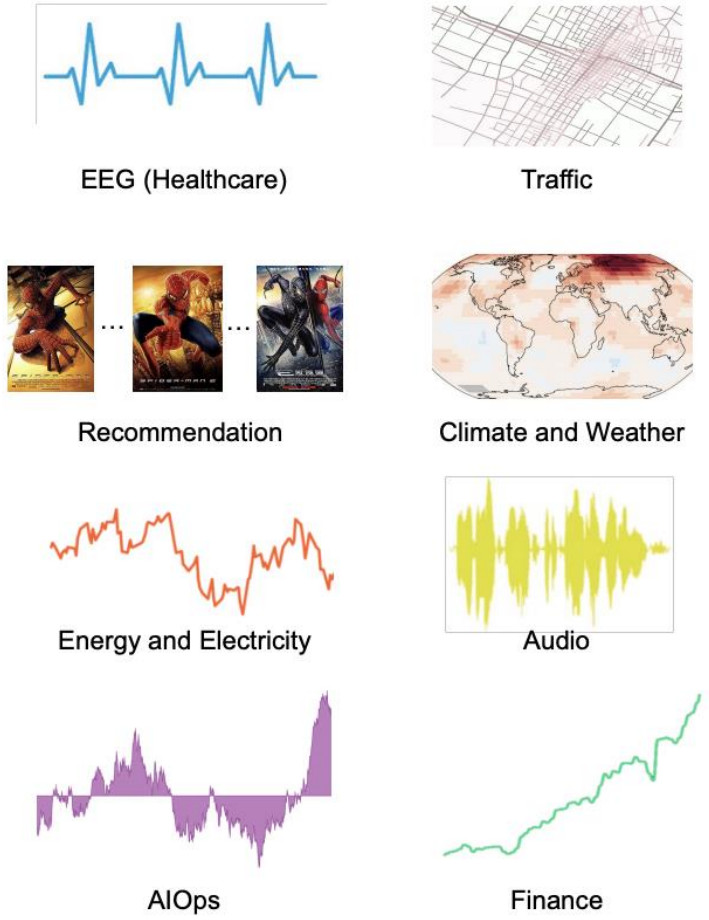


Fig. 8: Examples of time series and spatio-temporal data for different application scenarios.

References

- “What are Diffusion Models?” by Lilian Wang
- https://cs231n.stanford.edu/slides/2023/lecture_15.pdf
- CVPR 2022 tutorial: “Denoising Diffusion-based Generative Modeling: Foundations and Applications”
- “Denoising Diffusion Probabilistic Models”, Jonathan Ho, Ajay Jain, Pieter Abbeel 2022

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